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Tourism and livelihood sustainability in the Garhwal Himalaya

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Abstract

Tourism, particularly pilgrimage tourism, in the Garhwal Himalaya is a centuries-old practice. It contributes significantly to enhancing employment, income, and livelihood, as many people are engaged in it. This paper examines tourism practices in the Garhwal Himalaya and their role in livelihood sustainability. The study is empirical in nature. Data were collected through case studies of service centers located along the routes to major tourism destinations in the Garhwal Himalaya. A total of 200 heads of households were interviewed, including 100 who provided services to tourists/pilgrims and 100 who were engaged in traditional agriculture. Additional data on tourist/pilgrim inflow at the selected destinations were obtained from the Uttarakhand Tourism Development Board, Dehradun. The study reveals that income from tourism activities is considerably higher (over 40%) than income from traditional agriculture. It further suggests that tourism can significantly enhance livelihood sustainability if proper standard operating procedures are adopted.

Keywords Tourism, Pilgrimage, Tourist/pilgrim, Livelihood sustainability, Garhwal Himalaya

1 Introduction

Tourism is defined as the movement of people from one place to another for purposes such as leisure, peace, pilgrimage, or business, typically for a duration of six hours or more [1]. This movement may occur through trekking or by traveling via roadways, airways, or waterways, and encompasses various types of tourism. Every year, hundreds of thousands of tourists visit destinations of interest both within and outside their home countries [2]. Tourism comprises several components, primarily geographical and cultural. Among the geographical components, landscapes such as high mountains, river valleys, plains, and coastal regions are particularly significant [3]. Forested landscapes—including wildlife sanctuaries, national parks, biosphere reserves, and conservation areas—also form important geographical elements. Cultural components include pilgrimage sites, fairs, festivals, and the performance of traditional customs and rituals [4].

The Earth possesses rich diversity in both the geographical and cultural components of tourism, extending from the North Pole to the South Pole and from the Eastern Hemisphere to the Western Hemisphere. The world's mountains, river valleys, plateaus,



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plains, coastal regions, and landforms shaped by glaciers, rivers, and winds are invaluable treasures for tourism [5]. Cultural diversity across the globe—from the Americas to Africa, Europe, Asia, and Oceania—further promotes the growth of cultural tourism. Iconic destinations such as Kailash Mansarovar in the Himalaya, the Pyramids of Giza in Egypt, Machu Picchu in Peru, the ancient Mesopotamian region of the Euphrates and Tigris rivers, the Ganges Valley in India, and the Hanging Gardens of Babylon are among the major attractions for cultural tourism [6].

Tourism, often referred to as a smokeless industry, is one of the fastest-growing industries in the world. It contributes approximately 10% (USD 10.9 trillion) to the global GDP, although this figure varies from country to country depending on the level of tourism development [7]. According to the World Economic Forum [8], the world's top ten tourism destinations are the United States, Spain, Japan, France, Australia, Germany, the United Kingdom, China, Italy, and Switzerland. In countries such as Switzerland, Croatia, and Ireland, the economy and livelihoods are heavily dependent on tourism activities [9]. The tourism industry encompasses transportation, accommodation, and promotion, all of which require significant human capital. Therefore, its impact on livelihood sustainability is substantial. In urban centers, livelihoods are largely dependent on business and service sectors, including hospitality and transport, whereas rural areas rely primarily on agriculture and its allied practices [10]. Tourism practices, however, can play a crucial role in enhancing livelihoods in rural areas.

The Himalayan region is rich in both geographical and cultural components of tourism. The mighty Himalaya features highly elevated mountain ranges, low-lying river valleys, and foothill plains [11]. These physiographic features give rise to diverse landforms shaped by glaciers and rivers. The forest landscape varies with elevation, ranging from monsoon deciduous forests at lower altitudes to pine, mixed oak, deodar, and alpine pastures at higher altitudes [12]. Numerous other locations of tourist interest contribute to making the Himalaya one of the world's major tourist hotspots [13]. Its cultural richness further enhances its appeal as a tourism destination. The region is home to numerous highland and river-valley pilgrimage sites and sacred places that attract pilgrims from both within India and abroad in search of spiritual solace and peace of mind. Cultural tourism in the Himalaya is marked by religious processions, pilgrimages to high-altitude shrines, and the celebration of traditional fairs and festivals.

The Himalaya is the youngest and tallest mountain range in the world, where a significant portion of the population lives in poverty and below the poverty line [14]. The cultivation of traditional cereals is the primary occupation and main source of income, though it is often insufficient to sustain the growing population. Tourism, by contrast, holds immense potential to enhance livelihoods and achieve livelihood sustainability [15–18]. By sustainably harnessing the region's natural and cultural components, the Himalaya can serve as a unique destination for tourists and pilgrims, with tourism practices offering employment and income opportunities to a large segment of the population [19–23]. While a substantial volume of literature exists on tourism practices in the Himalaya, limited research has been conducted on the relationship between tourism and sustainable livelihoods in the region. This study aims to fill that gap and support future research in similar mountainous areas. The article examines tourism practices and their impact on livelihood sustainability in the Himalaya. It also explores how livelihood sustainability can be achieved by promoting sustainable tourism. The central hypothesis

is that sustainable livelihoods can be attained through increased community involvement in tourism development and improvements in infrastructural facilities at tourism destinations.

2 Literature review

Studies on tourism and related aspects in the Himalaya are extensive and abundant. Many scholars, both within and outside the country, have been working on this subject for a long time. However, studies linking tourism with rural livelihood sustainability are relatively few and limited. This study addresses this gap and provides insight into sustainable tourism for livelihood enhancement. In this paper, a selection of important and recent studies is reviewed to understand the current state of tourism development and its role in livelihood sustainability.

The present author [24] has conducted several studies on tourism and livelihood sustainability. He surveyed five villages in the Kali Kumaon region of the Uttarakhand Himalaya. The study found that tourism, among other income sources, holds strong potential to enhance livelihoods in high-altitude communities of Uttarakhand. Sati & Banerjee [25, 26] published their research on *Sustainable Homestay Tourism for Enhancing Rural Livelihood in the Tons River Basin in the Uttarakhand Himalaya*, illustrating that providing homestay tourism services along major trekking and mountaineering routes in the Himalaya can enhance rural sustainability. Pandey and Bhatt [27] assessed rural livelihood sustainability in the Champawat and Almora districts of Uttarakhand. They stated that sustainable rural tourism can generate employment and income, thereby improving livelihoods. Eco-tourism in the Himalayan region is one of the important forms of tourism for both economic development and environmental conservation. Balodi et al. [28] elaborated on how eco-tourism enhances income and reduces youth migration. Another study on the trends and potential of eco-tourism development in the Uttarakhand Himalaya was conducted by Sati [16, 18]. In this work, he analyzed eco-tourist inflows in national parks and wildlife sanctuaries and recommended infrastructure and policy interventions. Joshi and Sharma [29] conducted a study on social entrepreneurship through community-based eco-tourism policy in the Himalayan region, emphasizing the need for low-impact, sustainable tourism policies in the fragile Himalayan ecosystems. Kahkasha & Sharma [30] studied the design of AI solutions for sustainable tourism in Uttarakhand. They described how AI tools for environmental monitoring and tourist-flow management can empower local communities to act responsibly. A workshop report by RGICS [31] highlighted conservation-linked entrepreneurship and sustainable development in the mountain districts of Uttarakhand. It emphasized that eco-tourism, organic farming, and value addition linked to conservation can enhance livelihoods. The Government of Uttarakhand has initiated homestay tourism programs in rural areas. In Khati and Wachham villages, each family earned approximately ₹90,000 in just two months [32, 33]. A report from the state government stated that the state is promoting eco-tourism near forest zones and establishing four new eco-tourism zones to support sustainable livelihoods and conservation [32, 33]. This initiative aims to promote eco-tourism by enabling winter access to Gangotri National Park for snow leopard sightings, thereby boosting local employment and conserving biodiversity [34].

3 Methodology

3.1 The study area

The Garhwal Himalaya lies in the western part of the Uttarakhand Himalaya, stretching between 29°31'9" N to 31°26'5" N and 77°33'5" E to 80°6'0" E, and covering a total area of 29,089 km². It comprises seven districts, most of which are mountainous, with a significant portion of the area under perpetual snow. Notably, the districts of Uttarkashi and Chamoli each have approximately 16% of their area covered by permanent snow, while Rudraprayag has a smaller snow-covered area. The Garhwal Himalaya is the origin of the Ganga river system—one of the major river systems in the world. The region boasts a spectacular natural and cultural landscape. It is also characterized by a unique forest ecosystem [16, 18]. Diverse landscapes such as river valleys, mid-altitude zones, highlands, alpine pastures, and the Greater Himalaya contribute to its scenic beauty. Panoramic landforms including river terraces, gorges, waterfalls, cascades, and rivulets enhance its physical appeal. The region is dotted with numerous hill stations, trekking routes, high-altitude pilgrimage sites, and sacred places in the river valleys—all of which serve as prominent destinations for both natural and cultural tourism (Fig. 1).

The economy of the Garhwal region largely depends on traditional agriculture, with approximately 70% of the population engaged in agricultural activities. However, agricultural output is minimal, resulting in widespread poverty, with many people living below the poverty line. In contrast, the Garhwal Himalaya is home to world-renowned tourist destinations that generate income and provide employment opportunities for the local population, thereby enhancing their livelihoods. This article examines the role of tourism in promoting livelihood sustainability and argues that, if developed sustainably, tourism can serve as a viable means for the local community to achieve long-term livelihood security.

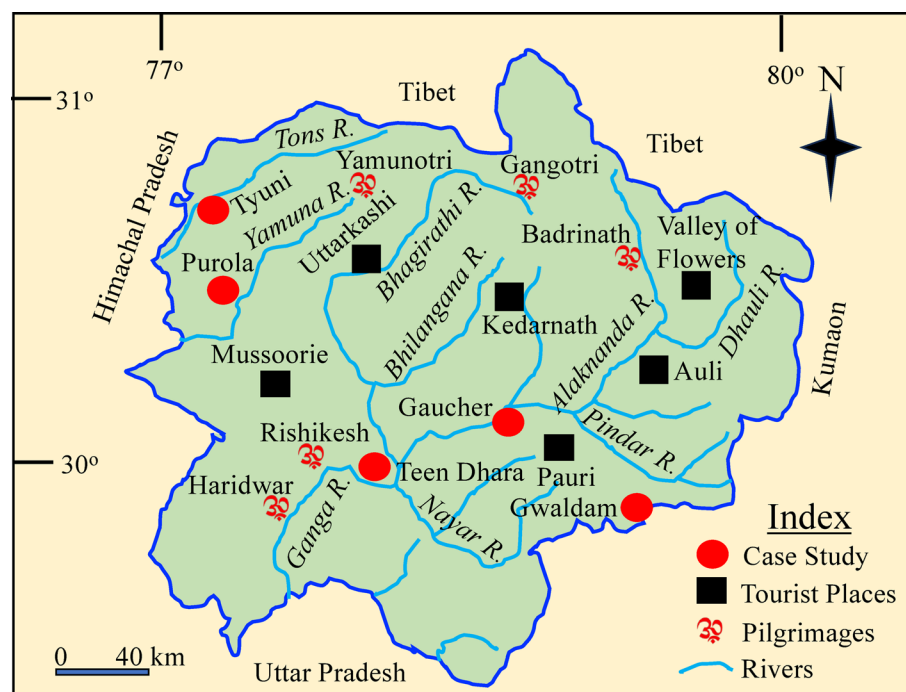


Fig. 1 Location map of Garhwal Himalaya and case studies service centres

3.2 Data collection and survey methods

This study employed a qualitative approach, and empirical data were collected through interviews with individuals involved in tourism and agriculture. An extensive field visit was conducted in July 2024, starting from Teen Dhara and ending in Gwaldam in the Pindar River valley, and Tyuni in the Tons River valley. The author stayed for two days in each service centre. Each centre had two types of economic backgrounds—tourism-dominated and agriculture-dominated—although all centres were involved in both providing services to tourists and pilgrims, and practicing agriculture. However, the proportion of these activities varied from one centre to another. A random sampling method was used to survey the owners of each enterprise. The author also applied the observation method by personally visiting the selected service centres. Five service centres were chosen for the study—Teen Dhara, Tyuni, Purola, Gwaldam, and Gaucher—located at altitudes ranging from 800 m (lowest) to 1950 m (highest), across the districts of Tehri, Dehradun, Uttarkashi, and Chamoli. These service centres are situated in various river valleys: the Ganga Valley, the Tons Valley, the Kamal Valley, the Pindar and Gomti Valleys, and the Alaknanda Valley (Table 1).

Teen Dhara primarily provides food and beverages to pilgrims from morning to evening, as it is located on the route to the famous highland pilgrimage site of Badrinath. Tyuni, a rural-based service centre, lies enroute to the trekking destinations of Hari Ki Doon and Kedarkantha, offering accommodation and food to trekkers and tourists. Purola, a rural town, is situated off the main route to the Yamunotri pilgrimage. Gwaldam, a semi-urban service centre located in the transitional zone between the Garhwal and Kumaon regions, offers accommodation and food to visitors. Lastly, Gaucher is a fully urban service centre with a wide range of accommodations and restaurants. Positioned on the route to another major highland pilgrimage site, thousands of pilgrims and tourists pass through Gaucher each year.

A total of 200 individuals were interviewed, including 100 engaged in tourism and 100 in agriculture. The respondents were randomly selected, considering their education and age. Almost all had completed high school or higher, and their average age was 40 years. They were engaged in both practicing agriculture and providing services to tourists and pilgrims. However, the number of respondents from each category varied across service centres, depending on their geographical location and economic base. For instance, service centres located in rural areas had economies more reliant on agriculture, and

Table 1 Case study service centres, their location, and characteristics.

Source: Field visits

Service centres	Altitude (m)	River valleys	Districts	Characteristics
Gaucher	800	Alaknanda	Chamoli	Gaucher, a wide and fertile valley of the Alaknanda River, is located on the way to Badrinath
Tyuni	950	Tons	Dehradun	Tyuni is semi-urban, located on the way to Hari-ki-Doon and Kedar Kantha trekking routes. It provides restaurants and accommodations
Teen Dhara	1475	Ganga	Tehri	Located on the hilltop on the way to Badrinath, it has many restaurants, servicing fresh food all the time
Purola	1524	Kamal	Uttarkashi	Purola is a rural service centre, located at a juncture of Tons and Yamuna River valleys
Gwaldam	1940	Pindar and Gomti	Chamoli	Gwaldam, a semi-urban centre, is located on the water parting of the Pindar and Gomti rivers and dividing culture of Garhwal and Kumaon regions

therefore, a greater number of people were involved in agricultural activities. Conversely, urban service centres had more individuals engaged in tourism-related services. The author employed a random sampling method to select respondents and gather data on the income earned from tourism and agriculture. The collected data were entered into an Excel sheet for analysis. Income figures were converted into USD for comparative purposes. The analysis revealed that income earned from tourism was significantly higher than that earned from agriculture. Additionally, the types of tourism services and the number of people involved in each were classified. Respondents' perceptions regarding tourism and livelihood sustainability were also recorded.

Perceptions of 120 people were obtained, and a total of 10 questions were asked to each person. The questions ranged from tourism's contribution to income share to the role of tourism in livelihood sustainability, availability of sufficient accommodation, adequacy of transportation facilities, government support for tourism development, the prominence of pilgrimage tourism as a primary form of tourism, tourism as a major source of livelihood, the role of homestay tourism, the role of people's participation in tourism development, and the serving of local food and beverages to tourists and pilgrims. In most cases, the respondents' positive responses exceeded 60%. However, they were 50% or lower in terms of adequate transportation facilities and government support for tourism development.

Data were also gathered from secondary sources, primarily from the Uttarakhand Tourism Development Board (UTDB), Dehradun, regarding tourist and pilgrim inflow to six selected tourist destinations—Uttarkashi, Pauri, Mussoorie, Auli, Kedarnath, and the Valley of Flowers. Additionally, data on the number of pilgrims who visited the Kedarnath highland pilgrimage during a 48-day period in 2025 were also obtained from UTDB. These data were presented in tables and illustrated through graphs for better understanding and analysis.

4 Results

4.1 Tourism in the Garhwal Himalaya

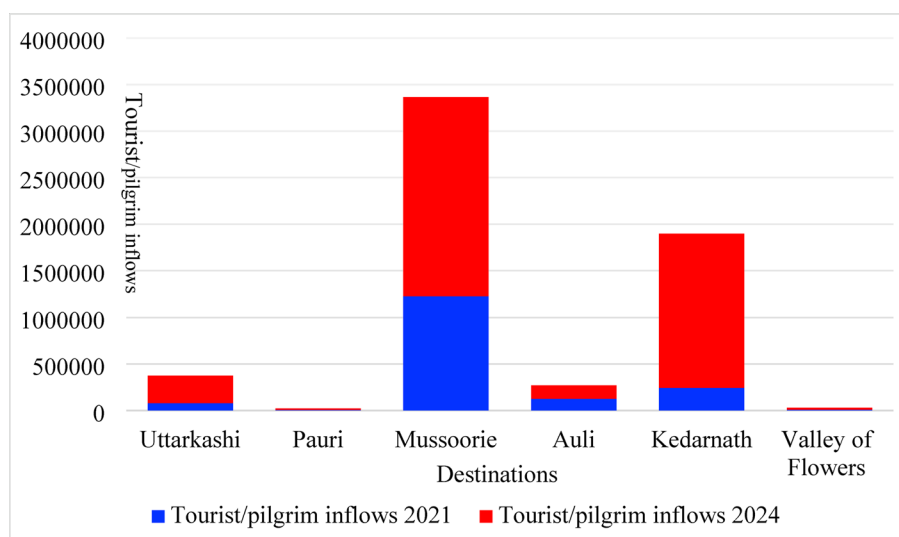
The Garhwal Himalaya is rich in both cultural and natural tourism destinations, attracting thousands of pilgrims and tourists each year. The major types of tourism practiced in the region can be broadly classified into four categories: cultural tourism, natural tourism, adventure tourism, and eco-tourism. For this study, the author selected six prominent tourist destinations from the Garhwal Himalaya. Of these, two are cultural tourism destinations—Uttarkashi and Kedarnath; two are adventure tourism destinations—Auli and the Valley of Flowers; one is a natural tourism destination—Mussoorie; and one is categorized as an administrative tourism destination—Pauri (Table 2). The altitudes of these destinations range from 1160 to 3600 m

Tourist and pilgrim inflow data for the years 2021 and 2024 were collected from the Uttarakhand Tourism Development Board (UTDB), Dehradun. The year 2021, affected by the COVID-19 pandemic, witnessed significantly lower tourist inflows, whereas 2024 showed a return to normalcy with a marked increase in visitors. The study reveals that Mussoorie recorded the highest tourist inflow in both years. Known as the “Queen of the Hills,” Mussoorie enjoys a pleasant summer climate and occasional snowfall in winter, making it a year-round destination. Tourist numbers peak during the summer when central India, including the Ganges Plain, experiences extreme heat.

Table 2 Tourists/pilgrims' inflows in the selected tourism destinations.

Source: UTDB, 2024. Altitude is the average of lowest and highest points of the tourist destinations

Destinations	Altitude (m)	Tourism type	Tourist/pilgrim inflows	
			2021	2024
Uttarkashi	1160	Cultural	80,733	294,590
Pauri	1770	Administrative	7350	16,089
Mussoorie	2000	Natural	1,229,808	2,134,626
Auli	2800	Adventure	127,488	142,227
Kedarnath	3580	Cultural	243,012	1,653,581
Valley of Flowers	3600	Adventure	10,398	19,436

**Fig. 2** Tourist/pilgrim inflows in 2021 and 2024 in the tourism destination of Garhwal

Kedarnath, a high-altitude pilgrimage site, is open for only six months during the summer. Despite its seasonal accessibility, it attracts a large number of pilgrims from both within and outside India. A remarkable 265% increase in pilgrim numbers was recorded over the study period. Uttarkashi, located on the banks of the Bhagirathi River, serves as a cultural hub and a transit point enroute to the Gangotri pilgrimage. While most pilgrims proceed directly to Gangotri, a smaller number visit the Kashi Vishwanath Temple in Uttarkashi, resulting in relatively modest footfall.

Pauri, an administrative city and the district headquarters, sees limited tourist activity, mostly from travellers enroute to Badrinath. Auli and the Valley of Flowers are located in the high-altitude Chamoli district. Auli is accessible by road and attracts adventure tourists, especially during the winter sports season. In contrast, the Valley of Flowers is remotely located and accessible only by trekking, thus attracting a smaller number of visitors. Nonetheless, an overall increase in tourist and pilgrim inflows was observed across all selected destinations.

Figure 2 depicts tourist and pilgrim inflows in 2021 and 2024 across the selected tourism destinations. It is evident that Mussoorie and Kedarnath received the highest number of tourists and pilgrims, respectively. A significant overall increase in tourist and pilgrim numbers was observed during this period. Notably, Uttarkashi recorded a substantial rise in tourist/pilgrim inflow, reflecting growing interest in cultural and pilgrimage tourism in the region.

Data on tourist and pilgrim inflows in 2021 and 2024 across the selected tourism destinations show a significant overall increase (Fig. 3). The highest growth in inflow was recorded in Kedarnath, followed by Uttarkashi. Kedarnath, one of the most revered high-altitude pilgrimage sites, is believed by devotees to offer liberation from the cycle of birth and death upon a single visit. Although Kedarnath is located in a remote area and requires a 16-km trek, it continues to attract a large number of pilgrims each year. Helicopter services are also available, making access easier for some visitors.

Uttarkashi, situated on the banks of the Bhagirathi River enroute to the Gangotri highland pilgrimage, is a prominent destination for cultural tourism. It recorded the second-highest increase in pilgrim inflow during the period. Pauri, a scenic administrative town, also witnessed a notable rise in tourist numbers. Mussoorie, a popular hill station known for its pleasant summer climate, attracts tourists year-round and experienced substantial growth in visitor numbers.

The Valley of Flowers, a UNESCO World Heritage Site accessible only by trekking, registered a steady increase in the number of trekkers. In contrast, Auli, known for its winter sports and adventure tourism, recorded only a slight increase in tourist inflow during the same period.

4.2 Pilgrims visited Kedarnath pilgrimage and income earned

According to the 2025 report by the Uttarakhand Tourism Development Board (UTDB) on pilgrimage tourism, approximately 3.2 million pilgrims visited the Char Dham shrines—Badrinath, Kedarnath, Gangotri, and Yamunotri—during the 48-day yatra. This section focuses on the Kedarnath pilgrimage. The yatra began on May 2, 2025, and by June 18, around 1.14 million pilgrims had visited Kedarnath, generating a total income of approximately USD 35,294,117. Pilgrims reached Kedarnath by trekking, helicopter, horses and ponies, and using Dandi and Kandi (wooden palanquins). Helicopter services generated about USD 7,058,823. There are over 2,000 hotels and restaurants along the yatra route, with around 350 located in Gaurikund alone. The average income from each pilgrim per day, including accommodation and food, was about USD 23.50, totaling approximately USD 17,647,059. Additional income of around USD 7,882,353 came from the use of horses and ponies, while USD 270,588 was earned from services using

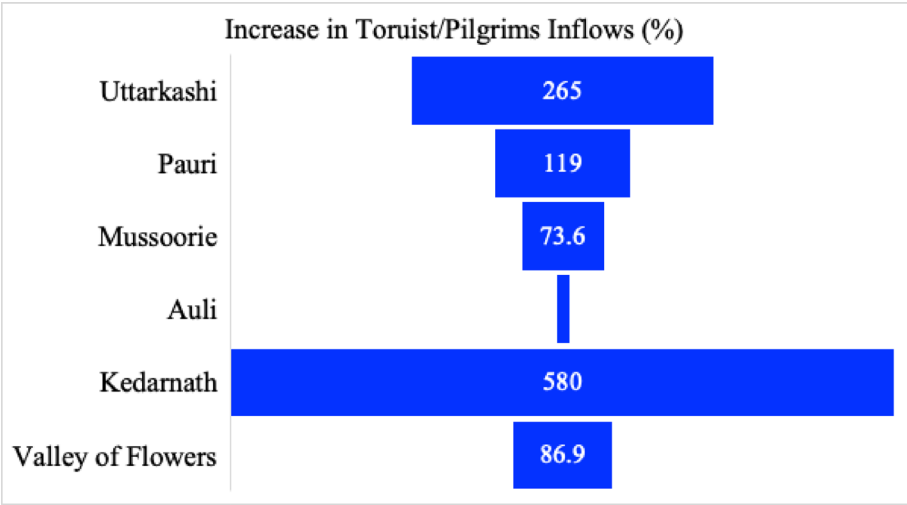


Fig. 3 Increase (%) in tourist/pilgrim inflow in the tourism destination from 2021 to 2024

Table 3 Pilgrims visited Kedarnath pilgrimage and income earned (May 2 to June 18).

Source: UTDB, 19 June 2025

S. No	Variables	Number	Income (USD)
1	Total pilgrims visited Kedarnath	1,140,000	35,294,117
2	Total number of hotels and restaurants	2000	17,647,059
3	Pilgrims travelled by helicopter	49,247	7,058,823
4	Pilgrims used Dandi and Kandi (total 7000)	10,000	270,588
5	Pilgrims used ponies and horses	227,614	7,882,353
6	225 shuttle buses (Sonprayag to Gaurikund)	1,140,000	1.1

1 USD = 85 INR (June 6, 2025)

Table 4 People involved in tourism and agricultural activities and average annual income (USD).

Source: Primary data collection

Service centres	People involved in tourism			People involved in agriculture		
	Number	Income	PCI	Number	Income	PCI
Gaucher	20	101,647.08	5082.35	20	50,823.48	2541.17
Tyuni	20	67,765.2	3388.26	20	50,823.48	2541.17
Teen Dhara	20	84,705.84	4235.29	20	42,352.92	2117.65
Purola	20	59,294.16	2964.71	20	59,294.16	2964.71
Gwaldam	20	59,294.16	2964.71	20	42,352.92	2117.65
Total	100	372,705.84	3727.06	100	245,646.96	2456.47

1 USD = 85 INR (June 6, 2025); PCI = Per Capita Income

Dandi and Kandi. On average, 24,000 pilgrims visited Kedarnath daily. Furthermore, 200 shuttle buses operated between Sonprayag and Gaurikund, each charging USD 1.10 per pilgrim.

Along with the increasing number of pilgrims visiting the Kedarnath highland pilgrimage, several challenges have emerged, particularly concerning the safety of pilgrims. During this period, three helicopter crashes occurred, resulting in the deaths of eight people. The entire route from Rishikesh to Kedarnath is fragile and highly vulnerable to landslide hazards. The stretch of road between Rudraprayag and Gaurikund is especially prone to landslides. Additionally, the approximately 19-km trek from Gaurikund to Kedarnath is steep and precipitous. Along the route, several streams become extremely violent during the rainy season, posing serious risks and leading to fatalities. So far, around 10 pilgrims have lost their lives due to landslides and flash floods, in addition to those who died in helicopter crashes (Table 3).

4.3 Income from tourism and agricultural activities

Five service centres were surveyed for this study (Table 4). Residents in these centres are engaged in primary activities such as agriculture and livestock farming, and they also provide services to tourists and pilgrims. From each service centre, 40 individuals were interviewed—20 involved in agriculture and 20 in tourism-related activities. The income from each activity was calculated annually and converted into USD at the exchange rate of 1 USD = 85 INR. The total annual income generated from tourism activities across the five centres was USD 372,705.84, with Gaucher reporting the highest income and Purola and Gwaldam the lowest.

The higher income in Gaucher and Teen Dhara is attributed to their strategic locations along the main routes to the prominent pilgrimage destinations of Badrinath and Kedarnath. Teen Dhara primarily offers food and beverages and is recognized as a key stop for

pilgrims, while Gaucher provides a substantial range of quality accommodations. In contrast, Purola and Gwaldam have a rural character and are located off the main pilgrimage routes, which contributes to their lower tourism-related income. Per capita income (PCI) from tourism also varies accordingly across these centres.

In terms of average annual income earned from agricultural activities, the total across the five service centres was USD 245,646.96, which was considerably lower than the income generated from tourism. The income varied from USD 59,294.16 in Purola to USD 42,352.92 each in Teen Dhara and Gwaldam. Correspondingly, the per capita income (PCI) also varied. It was observed that the service centres of Purola and Tyuni have relatively large areas of fertile arable land and practice intensive agriculture, which contributes to higher agricultural income compared to the other centres. However, when compared to tourism, the income from agriculture remains significantly lower.

4.4 Comparison of income from tourism and agriculture

A comparison of income between tourism and agricultural activities reveals that tourism is significantly more profitable, with an overall income difference of +36.36%. The highest differences were observed in Teen Dhara and Gaucher, where income from tourism exceeded that from agriculture by 50% each. In Purola, the difference was relatively modest at +14.29%, followed by Tyuni at +25% and Gwaldam at +28.57%. The data indicate that service centres with a rural background show a smaller income gap between tourism and agriculture, suggesting that agriculture remains a strong livelihood base in these areas (Fig. 4).

4.5 Classification of tourism services and people involved

In the Garhwal Himalaya, local people offer a variety of services to tourists and pilgrims, ranging from food and beverage services to accommodation and transportation. Transportation services include buses, taxis, ponies, and Dolies (palanquins). Figure 5 illustrates the proportion of different types of services provided in the surveyed service centres. Among the major services, restaurant-related activities have the highest level of

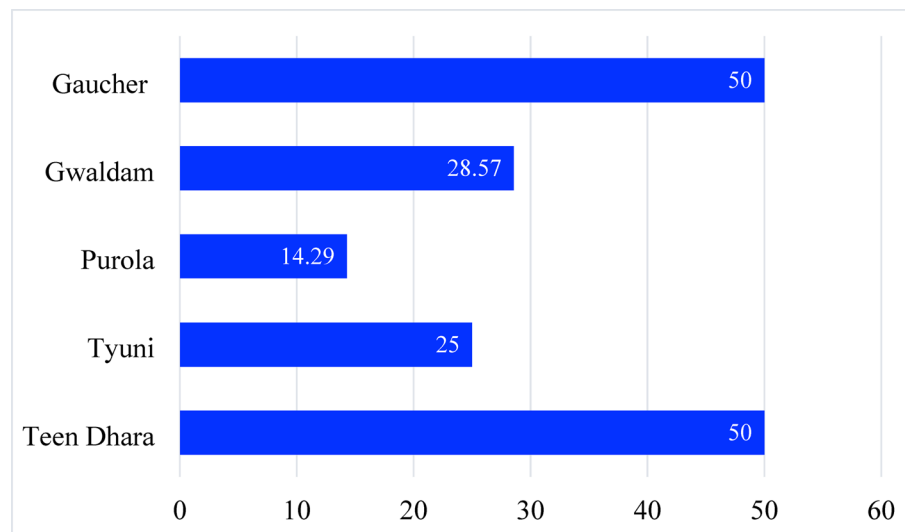


Fig. 4 Percentage of increased income earned from practicing tourism in comparison to agricultural practices in the selected service centres

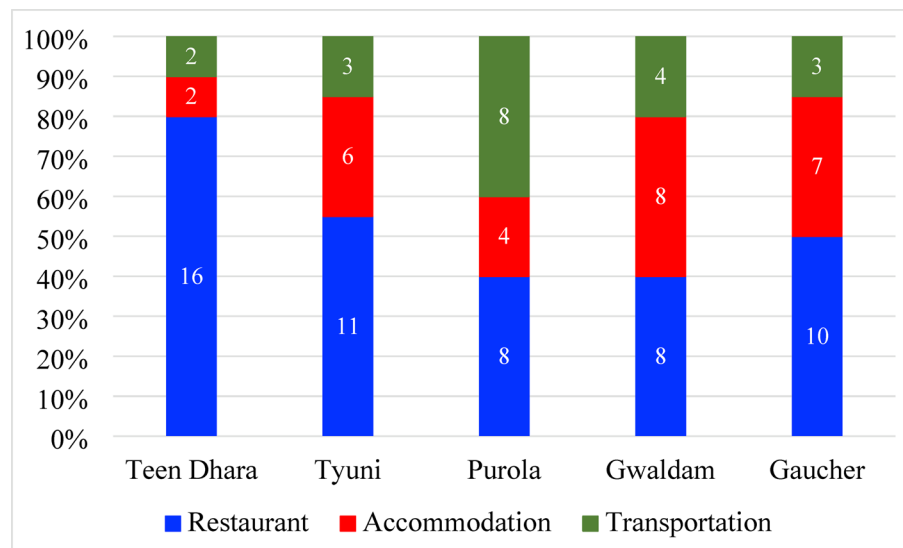


Fig. 5 Classification of tourism services and people involved

involvement, followed by accommodation services. Only a small number of people are engaged in providing transportation.

However, the proportion of involvement in various services varies across the service centres. For instance, approximately 80% of people are involved in restaurant services overall, but this figure drops to just 40% in Purola and Gwaldam. In Gwaldam (40%) and Gaucher (35%), a significant portion of the population is engaged in providing accommodation services. Conversely, in Purola, a large number of people are involved in offering transportation services to tourists and pilgrims. These variations indicate that the location and accessibility of each service centre largely determine the nature and type of services provided.

4.6 Tourism and livelihood sustainability

Tourism in the Himalayan region has emerged as a significant economic activity with substantial potential to enhance livelihoods and promote sustainability. It is a multi-faceted sector that not only generates income and employment but also helps preserve and strengthen local culture and cultural practices. As a livelihood option, sustainable tourism supports local socio-economic systems and is comparatively less vulnerable to environmental degradation and market volatility [35]. In the Central Himalayan region, tourism primarily takes the form of pilgrimage to highland sacred sites such as the Char Dhams, Shakti Peeths, and Jyotirlingas. Additionally, nature-based tourism is prominent in destinations like the Valley of Flowers, Govind Wildlife Sanctuary, and Gangotri National Park. These forms of tourism are marked by seasonal influxes of visitors. Overall, tourism in the Himalayan region has a strong multiplier effect on both the local and regional economies [22, 23].

Community-based and homestay tourism is emerging as a vital livelihood support system for local people in the Himalayan region [25, 26]. It not only generates income but also serves as a platform to showcase cultural heritage and traditional local cuisines, including food and beverages. The integration of tourism with local culture contributes to the conservation of indigenous knowledge systems and promotes community

empowerment [36]. Sustainable tourism can be achieved through effective management and carrying capacity analysis of tourist destinations. Such practices help mitigate the challenges of over-tourism and environmental degradation. However, in remote tourism destinations, poor connectivity, lack of capacity-building programs, and limited health and sanitation facilities continue to hinder tourism development [37]. Addressing these challenges is essential for ensuring long-term livelihood sustainability.

4.7 People's perception

A total of 120 heads of households were interviewed to understand their perceptions of tourism practices and livelihood sustainability (Table 5). Ten questions were asked, covering various aspects of tourism, and the responses were recorded in terms of frequency and percentage. About 65% of respondents believed that tourism provides the highest income among available livelihood options. A significant number of respondents also perceived that tourism has the potential to ensure livelihood sustainability.

When asked about the sufficiency of accommodation facilities, only half of the respondents responded positively. A similar trend was observed regarding transportation facilities, indicating that both accommodation and transportation infrastructure are currently inadequate. Less than 50% of respondents believed that the government is providing sufficient support for tourism development. This highlights a perceived lack of effective governmental intervention and infrastructure support in the tourism sector.

Respondents identified pilgrimage tourism as the primary form of tourism in the Garhwal region. Overall, there was a general perception that tourism can become a major source of livelihood. Additionally, community-based homestay tourism, increased people's participation in tourism development, and offering local food and beverages to tourists and pilgrims were seen as key strategies to further improve local livelihoods.

4.8 Discussion and implications

Tourism in the Himalaya has been recognized as a key driver of sustainable rural livelihoods. Previous studies have shown that tourism plays a significant role in enhancing socio-economic resilience in mountain regions [38, 39]. The contribution of tourism to income generation and livelihood sustainability in mountain regions was assessed

Table 5 People's perception of tourism practices and livelihood sustainability (n = 120).

Source: By author

S. No	Questions	Frequency	Frequency (%)
1	Does tourism contribute the highest share of income?	78	65
2	Can tourism ensure livelihood sustainability?	72	60
3	Is accommodation sufficient at tourism destinations?	64	53.3
4	Are transportation facilities adequate?	60	50
5	Does the state government provide support for tourism development?	53	44.2
6	Is pilgrimage tourism the primary form of tourism in the Garhwal region?	92	76.7
7	Do you think tourism can become a major source of livelihood?	70	58.3
8	Do you believe that homestay tourism can enhance livelihood opportunities?	95	79.2
9	Can people's participation contribute to the development of tourism?	72	60
10	Can serving local food and beverages to tourists improve livelihood prospects?	86	71.7

through theoretical frameworks such as the sustainable livelihood approach and community-based tourism [40]. Several empirical studies in the Himalayan region depict that tourism reduces dependency on subsistence agriculture and improves infrastructural facilities [41]. Integrated tourism planning that harmonizes conservation and livelihood enhancement is essential. It can promote capacity building among local communities and establish policy mechanisms for equitable benefit-sharing. Strengthening institutional linkages and promoting eco-cultural tourism can further enhance livelihood sustainability.

In the Himalaya, various types of tourism are prevalent—such as natural, adventure, and cultural. Cultural tourism, particularly in the form of highland pilgrimage visits, has been a tradition for centuries. Pilgrimage tourism is mainly practiced during the six months of the summer season, from mid-April to mid-October, commonly referred to as the *Yatra* season. During this period, a large number of pilgrims visit sacred highland destinations. Along the pilgrimage routes, local people provide essential services such as accommodation and food and beverage offerings to the pilgrims. In earlier times, when pilgrims trekked to these remote shrines, they largely managed their own food and shelter, spending minimal money. Over time, with the construction of roads to many highland pilgrimage sites, accessibility has significantly improved. Pilgrims now often travel by public or private vehicles and stay at established service centres or designated tourist destinations. This shift has contributed to increased income for local communities and boosted the regional economy, benefiting both residents and government revenues.

Natural and adventure tourism in the Garhwal Himalayan region is steadily developing. The region is home to several world-famous hill towns such as Mussoorie, Nainital, Almora, and Ranikhet, which attract a large number of tourists throughout the year. Adventure tourism activities such as river rafting, trekking, and mountaineering are widely practiced and contribute significantly to employment generation and income enhancement. Towns like Rishikesh, Kadiyala, Mori, and Tanakpur have emerged as major centres for river rafting. Similarly, trekking is popular in destinations such as Hari Ki Doon, Kedar Kantha, Satopanth, Panchachuli, the Valley of Flowers, and Adi Kailash. Mountaineering is also gaining popularity in the Himalayan region, with expeditions to prominent peaks becoming an emerging trend. Collectively, these tourism activities play a crucial role in promoting livelihood sustainability for local communities in the Himalayan region.

Figure 6 illustrates the relationship between tourism and livelihood sustainability in the Garhwal Himalaya. The region primarily practices three types of tourism—pilgrimage, natural, and adventure—all of which are in a developed stage. Tourism plays a vital role in enhancing livelihood sustainability, as it generates employment, increases income, and promotes the consumption and appreciation of local food and beverages. The figure also highlights that livelihood sustainability can be effectively achieved if certain measures are implemented. These include adopting standard operating procedures for helicopter services, regulating the influx of tourists and pilgrims, making registration mandatory for visitors to tourist destinations, and ensuring the availability of adequate infrastructural facilities. With these practices in place, tourism can continue to serve as a sustainable livelihood option for the people of the Garhwal Himalaya.

The findings of our empirical study reveal that the income earned from tourism was nearly double that of agriculture. Similarly, the per capita income from tourism-related

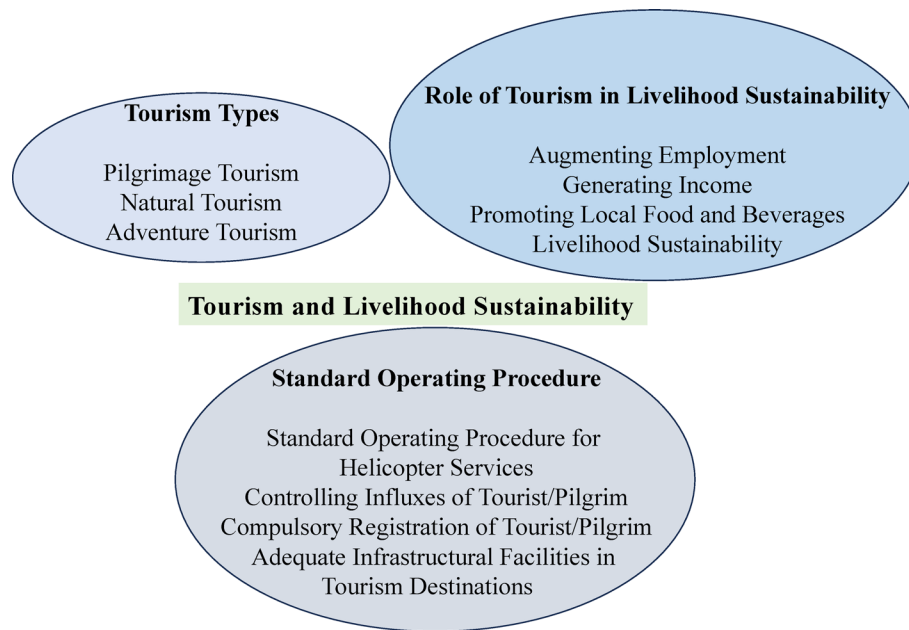


Fig. 6 Tourism and livelihood sustainability

activities was significantly higher. Although income from tourism was relatively lower in rural-based service centres, it still exceeded the income generated from agricultural activities. Overall, tourism income across the selected destinations was consistently higher than agricultural income. These findings suggest that tourism has greater potential to generate income compared to agriculture, especially in areas endowed with rich geographical and cultural diversity. Therefore, in such destinations, tourism can serve as a more effective and sustainable livelihood option than traditional agricultural practices.

The perception of interviewees involved in tourism-related activities revealed that tourism in the Himalaya holds high potential for livelihood sustainability and is emerging as a major source of income and employment. Although they acknowledged that tourism infrastructure is currently inadequate, they strongly believed that tourism can serve as a viable and sustainable livelihood option. It has already contributed significantly to the state's economy. According to the respondents, the majority of tourists visiting Uttarakhand travel to pilgrimage sites. They expressed that if adequate transportation and accommodation facilities are developed and quality food and beverages are consistently provided to tourists and pilgrims, it would further enhance both individual income and the regional economy.

It has been observed that highland pilgrimages in the Himalaya are extremely fragile and vulnerable to natural disasters. The *Yatra* season often coincides with the onset of the monsoon in India. The high intensity and frequency of monsoon rains frequently trigger cloudbursts, landslides, mass movements, flash floods, and debris flows. As a result, natural disasters occur almost every year, leading to the loss of lives among pilgrims and tourists. There are several recorded instances of such disasters in the past, which have significantly impacted tourism activities in the region. Tourist/pilgrim inflows tend to decrease when the intensity of natural disasters is high, and increase when conditions are safer. However, despite these risks, many pilgrims continue to visit highland shrines even in adverse conditions. This is largely driven by their deep-rooted

beliefs and traditions, which hold that visiting these sacred sites at least once in a life-time liberates the soul from the cycle of birth and death.

5 Conclusion

Tourism is one of the key livelihood options in the Garhwal Himalaya. It generates income, creates employment opportunities, contributes to livelihood sustainability, and reduce dependence on traditional subsistence systems. However, the region faces significant challenges due to natural disasters and the high influx of tourists and pilgrims, which can hinder tourism activities and impact local livelihoods. This study suggests that for sustainable tourism development and long-term livelihood sustainability, a well-defined Standard Operating Procedure (SOP) should be formulated and implemented to regulate tourist and pilgrim flow. Such measures would also aid in disaster risk reduction. Local communities should be actively involved in tourism and related activities to ensure inclusive development. Promoting local products—including traditional food and beverages—and serving them to tourists and pilgrims can further strengthen local economies. Additionally, improving tourism-related infrastructure will contribute to the growth of sustainable tourism and enhance livelihood opportunities in the region.

Meanwhile, this research is limited by seasonal variations in data and spatial coverage. It does not address the socio-ecological impacts of tourism. The qualitative perceptions of women and migrants require deeper exploration; therefore, future research should adopt longitudinal and mixed-method approaches to better assess the relationship between tourism, climate change, and livelihood adaptation. A comparative study across the entire Himalayan region is essential for effective policy formulation to achieve sustainable and regenerative mountain tourism.

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Author contributions

I am a single author and all works were done by me.

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Data availability

All data generated or analysed during this study are included in this published article.

Declarations

Ethics approval and consent to participate

This study involved 200 human participants, who were interviewed regarding their economic activities and income from each source, particularly in the agricultural sector and tourism. The study was approved by the Research Ethics Committee of Mizoram University, India. It is further certified that the study was conducted in accordance with the ethical standards laid down in the 1964 Declaration of Helsinki. Informed consent was obtained from all participants. For respondents under 18 years of age, consent was obtained from a parent and/or legal guardian.

Consent for publication

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Clinical trial

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References

- United Nations World Tourism Organization (UNWTO). International recommendations for tourism statistics. UNWTO. 2024. <https://www.unwto.org>.
- UNWTO. Tourism and culture synergies. World Tourism Organization. 2022. <https://www.unwto.org>.
- Mathieson A, Wall G. Tourism: economic, physical and social impacts. Longman; 1982.
- Singh S. Domestic tourism in India: trends and patterns. In: Singh S, editor. Domestic tourism in Asia: diversity and divergence. Earthscan; 2009. p. 69–90.
- Hall CM, Page SJ. The geography of tourism and recreation: environment, place and space. 4th ed. Routledge; 2014.
- Richards G. Cultural tourism: global and local perspectives. Routledge; 2007.
- WTTC. Travel and tourism economic impact research, London, UK. 2024. <https://wtcc.org/research/economic-impact>.
- WEF. Travel and tourism development index, World Economic Forum, New York. 2024. <https://www.weforum.org/publications/travel-tourism-development-index-2024>.
- Ashley C, Roe D, Goodwin H. Pro-poor tourism strategies: making tourism work for the poor. Overseas Development Institute; 2001.
- Briedenhann J, Wickens E. Tourism routes as a tool for the economic development of rural areas—vibrant hope or impossible dream? *Tour Manage*. 2004;25(1):71–9. [https://doi.org/10.1016/S0261-5177\(03\)00063-3](https://doi.org/10.1016/S0261-5177(03)00063-3).
- Sharma P. Tourism and livelihood in the Himalayas: challenges and opportunities. ICIMOD (International Centre for Integrated Mountain Development) 2000.
- Singh RB, Mishra D. Sustainable tourism and pilgrimage in the Indian Himalayas: opportunities and challenges. In: Singh RB, editor. The Himalayan environment: issues and concerns. New Delhi: Shree Publishers; 2004. p. 161–77.
- Godde PM, Price MF, Zimmermann FM (eds). Tourism and development in mountain regions. CABI Publishing; 2000.
- Jodha NS. Economic globalization and its repercussions for fragile mountain areas: a case from the Himalayas. In: Messerli B, Ives JD, editors. Mountains of the world: a global priority. The Parthenon Publishing Group; 2005. p. 71–82.
- Pandey B, Kumar R. Sustainable tourism and livelihood security in the Indian Himalayan Region. *J Mt Sci*. 2020;17(4):909–22. <https://doi.org/10.1007/s11629-019-5731-4>.
- Sati VP. Sustainable Tourism Development in the Himalaya: Constraints and Prospects. Springer; 2021.
- Sati V P. Pilgrimage tourism in Uttarakhand Himalaya: pilgrims' inflows and trends. *J Multidiscip Acad Tour*. 2023;8(2):109–17.
- Sati VP. Trends and potential of eco-tourism development in Uttarakhand Himalaya. *J Tour Sustain*. 2021;4(1):28–41.
- Mishra PK, Rout HB. Tourism and sustainable livelihood: a study of mountain communities in India. *Int J Sustain Dev World Ecol*. 2018;25(3):217–26. <https://doi.org/10.1080/13504509.2017.1408710>.
- Sati V P. Pilgrimage tourism in mountain regions: socio-economic implications in the Garhwal Himalaya. *South Asian J Tour Herit*. 2015;8(1):164–82.
- Sati VP. Carrying capacity analysis and destination development: a case study of Gangotri tourists/pilgrims' circuit in the Himalaya. *Asia Pac J Tourism Res*. 2018;23(3):312–22.
- Sati V P. The nature of tourism and tourists/pilgrims' inflow in Uttarakhand Himalaya. *J Multidiscip Acad Tour*. 2020;5(2):115–24.
- Sati VP. Sustainable tourism in the Himalaya: concept and practices. Springer; 2020.
- Sati VP. Exploring livelihood opportunities in the Kali Kumaon region of Uttarakhand Himalaya. *Afr J Rural Dev*. 2025;10(2):95–108.
- Sati VP, Banerjee A. Sustainable homestay tourism for enhancing rural livelihood in the Uttarakhand Himalaya: a study of the Tons River Basin. 2025.
- Sati VP, Banerjee A. Sustainable homestay tourism for enhancing rural livelihood in the Uttarakhand Himalaya: a study of the Tons River Basin. *BIO Web Conf*. 2025;83:03001.
- Pandeya P, Bhatt N. Assessing rural livelihood sustainability through field-based analysis: evidence from Champawat & Almora district of Uttarakhand. *Int J Res Innov Soc Sci*. 2025;9(6):202–10.
- Balodi G, Negi GCS, Thapliyal P, Bhatt M. Eco-tourism: a sustainable livelihood option for mountainous communities in Uttarakhand, India. *J Stud Dyn Change*. 2014;1(2):45–55.
- Joshi A, Sharma A. Social entrepreneurship through a community-based eco-tourism policy in the Himalayan state of Uttarakhand. *J Tour Hospit*. 2020;9(1):1000432.
- Kahkasha N, Sharma S. Designing AI solutions for sustainable tourism in Uttarakhand. *Int Educ J Sci Eng*. 2025;7(3):70–9.
- Rajiv Gandhi Institute for Contemporary Studies. Workshop report: conservation linked entrepreneurship and sustainable development of mountain districts of Uttarakhand. RGICS. 2023. Retrieved from <https://www.rgics.org/environment/workshop-report-conservation-linked-entrepreneurship-and-sustainable-development-of-mountain-districts-of-uttarakhand/>.
- Bhatt M. Rooted & thriving: 2 Bageshwar villages defy Uttarakhand's migration tide. *The Times of India*. 2025, July 6. Retrieved from <https://timesofindia.indiatimes.com/city/dehradun/rooted-thriving-2-bageshwar-villages-defy-uttarakhand-migration-tide/articleshow/121915290.cms>.
- Bhatt M. Uttarakhand pays Rs 19.5 crore to families affected by wildlife conflicts. *The Times of India*. 2025, June 28. Retrieved from <https://timesofindia.indiatimes.com/city/dehradun/uttarakhand-pays-rs-19-5-crore-to-families-affected-by-wildlife-conflicts/articleshow/121808529.cms>.
- Sharma P. Spot snow leopard: Uttarakhand plans year-round access to Gangotri National Park. *The Times of India*. 2025, July 17. Retrieved from <https://timesofindia.indiatimes.com/city/dehradun/uttarakhand-plans-to-keep-gangotri-park-open-in-winter-to-boost-snow-leopard-tourism/articleshow/122664270.cms>.
- Scheyvens R, Biddulph R. Inclusive tourism development. *Tour Geogr*. 2018;20(4):589–609.
- Tosun C. Limits to community participation in the tourism development process in developing countries. *Tour Manage*. 2000;21(6):613–33.
- Nepal SK. Tourism-induced rural energy consumption in the Annapurna region of Nepal. *Tour Manage*. 2008;29(1):89–100.
- Nepal SK. Mountain tourism and sustainable development: coping with environmental and social challenges. *J Sustain Tour*. 2020;28(8):1221–38.
- Singh RB, Mishra A. Tourism and livelihood sustainability in the Indian Himalaya: a regional development perspective. *Environ Dev Sustain*. 2021;23(6):9051–69.

40. Ashley C, Carney D. Sustainable livelihoods: lessons from early experience. Department for International Development (DFID); 1999.
41. Baniya R, Poudel K. An analysis of push and pull travel motivations of domestic tourists in Nepal. *J Manage Dev Stud*. 2016;27:16–30.

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